The Computational Convergence Theorem

A Geometric Theory of Compute Architecture Under Physical Constraints

Author Name
Affiliation
email@example.com

March 27, 2026

Abstract

As frequency scaling stalls, modern computing systems are increasingly constrained not by raw arithmetic throughput alone, but by the cost of moving data through space. This paper presents the *Computational Convergence Theorem*, a geometric framework for reasoning about compute architecture under bounded clockspeed, distance-dependent bandwidth, and volumetric capacity scaling. We model architecture design as a constrained spatial optimization problem over compute density. The resulting minimizers fall into three regimes: *convergent* (latency-optimized), *diffuse* (throughput-optimized), and *hybrid* (mixed core–shell forms). This formulation unifies architectural trends across CPUs, GPUs, chiplets, and stacked-memory systems, and provides a mathematical basis for understanding why, when frequency scaling stops, geometry becomes the architecture.

1 Introduction

For decades, performance growth in computing systems was driven by transistor scaling and frequency increases. As voltage scaling slowed and power density limits emerged, clockspeed plateaued, shifting the primary optimization problem from *time* to *space*. In contemporary systems, the performance bottleneck increasingly arises from communication distance, transport bandwidth, and data locality rather than arithmetic capability alone.

This paper proposes a geometric interpretation of architecture design: modern compute systems can be modeled as spatial configurations minimizing data movement cost under physical constraints. The central claim is that, once frequency scaling converges, the dominant design variable becomes geometry. We formalize this statement through a constrained cost functional over compute density and show that its minimizers correspond to three distinct architectural regimes.

Contribution. We provide:

(i) a **clock convergence law** linking transistor growth, power bounds, and propagation delay,

(ii) a **spatial cost functional** for architecture design,

(iii) a **regime theorem** yielding convergent, diffuse, and hybrid minimizers, and

(iv) an **empirical interpretation** connecting the model to CPUs, GPUs, chiplets, and near-memory systems.

2 Related Work

The framework here complements several established results in computer architecture and scaling theory:

- **Amdahl's Law** formalizes serial bottlenecks in parallel speedup.
- **Gustafson's Law** reframes scaling when workload size grows with processor count.
- The **memory wall** highlights the growing gap between processor speed and memory access.
- **Dennard scaling** and its breakdown explain why voltage and frequency no longer scale in lockstep with transistor density.
- The **Roofline model** characterizes performance bounds imposed by arithmetic intensity and bandwidth.

These works focus on workload scaling, algorithmic limits, or empirical performance bounds. The present framework differs by treating architecture itself as a *geometric minimization problem* over space.

3 Physical Constraints

We begin with three physical assumptions that are sufficiently general to outlive any specific manufacturing node.

3.1 Bounded Clockspeed

Clock frequency is limited by both power density and signal propagation:

$$f \leq f_{\max} = \min\left(\frac{P_{\max}}{\kappa C V^2 N}, \frac{v}{L}\right), \tag{1}$$

where:

- $P_{\max}$ is the maximum power budget,
- κ is an activity factor,
- C is effective capacitance per switching unit,
- V is supply voltage,
- N is transistor count,
- v is effective signal propagation velocity,
- L is a characteristic system diameter.

3.2 Distance-Dependent Bandwidth

Definition 1 (Bandwidth Scaling Law)**.** *For interconnect distance d, the sustainable bandwidth is modeled as*

$$B(d) = \frac{B_0}{d^k}, \qquad k \geq 1, \tag{2}$$

where B_0 is a technology-dependent constant and k captures transport penalty.

Typical interpretations:

- $k = 1$: propagation-dominated transport,
- $k = 2$: RC-limited transport.

3.3 Volumetric Capacity Scaling

Definition 2 (Capacity Scaling)**.** *For a region $\Omega \subset \mathbb{R}^3$,*

$$\mathrm{Cap}(\Omega) = \eta|\Omega|, \tag{3}$$

where η is effective capacity density.

This captures the intuition that memory, storage, and parallel resource budgets grow with physical extent.

4 Clock Convergence Law

Assume transistor count follows Moore-type exponential growth:

$$N(t) = N_0 e^{\gamma t}, \tag{4}$$

where $N_0 > 0$ and $\gamma > 0$ is an effective growth rate.

Dynamic power is modeled as

$$P(t) = \kappa C V^2 f(t) N(t). \tag{5}$$

Imposing the power ceiling $P(t) \leq P_{\max}$ gives

$$f(t) \leq \frac{P_{\max}}{\kappa C V^2 N(t)}. \tag{6}$$

Substituting **??**:

$$f(t) \leq \frac{P_{\max}}{\kappa C V^2 N_0 e^{\gamma t}}. \tag{7}$$

Let

$$K := \frac{P_{\max}}{\kappa C V^2 N_0}, \tag{8}$$

then

$$f(t) \leq K e^{-\gamma t}. \tag{9}$$

Propagation imposes the independent bound

$$f(t) \leq \frac{v}{L}. \tag{10}$$

Combining **????**:

$$\boxed{f(t) \leq \min\left(K e^{-\gamma t}, \frac{v}{L}\right)} \tag{11}$$

Lemma 1 (Frequency Evolution)**.** *Under* **????***, clock frequency is bounded by* **??***.*

Theorem 1 (Clock Convergence)**.** *As* $t \to \infty$*,*

$$\limsup_{t\to\infty} f(t) \leq \frac{v}{L}. \tag{12}$$

In the asymptotic regime where transistor growth continues but power remains bounded, frequency scaling converges to a geometry-dependent limit.

Remark 1. *The exponential term in* **??** *decays with transistor growth, whereas the propagation term depends only on geometry. Thus the asymptotic bound is determined by spatial extent.*

5 Spatial Optimization Model

Let $\Omega \subset \mathbb{R}^3$ denote the architectural domain and let

$$\rho(x) \geq 0 \tag{13}$$

be compute density at position $x \in \Omega$, with total compute budget

$$\int_\Omega \rho(x)\ \mathrm{d}x = N. \tag{14}$$

Define the Euclidean distance

$$d(x, y) = \|x - y\|. \tag{15}$$

The system-level communication cost is modeled as

$$\mathcal{T}[\rho] = \iint_{\Omega\times\Omega} \rho(x)\rho(y) \left[\alpha\, d(x, y) + \beta\, d(x, y)^k\right]\ \mathrm{d}x\ \mathrm{d}y, \tag{16}$$

where:

- $\alpha > 0$ weights latency sensitivity,
- $\beta > 0$ weights bandwidth/throughput sensitivity,
- $k \in [1, 2]$ is inherited from **??**.

The optimization problem is:

$$\rho^* = \arg\min_{\rho\geq 0} \mathcal{T}[\rho] \quad \text{subject to} \quad \int_\Omega \rho(x)\ \mathrm{d}x = N, \tag{17}$$

optionally with the additional pointwise bound

$$0 \leq \rho(x) \leq \rho_{\max} \tag{18}$$

to prevent nonphysical singular collapse.

6 Variational Condition

Introduce the constrained Lagrangian

$$\mathcal{L}[\rho] = \mathcal{T}[\rho] - \lambda \left(\int_\Omega \rho(x)\ \mathrm{d}x - N \right), \tag{19}$$

where λ is a Lagrange multiplier.

Let η be an admissible perturbation satisfying $\int_\Omega \eta(x)\ \mathrm{d}x = 0$. Then the first variation condition

$$\delta\mathcal{L}[\rho; \eta] = 0 \tag{20}$$

yields

$$2 \int_\Omega \rho(y) \left[\alpha\, d(x,y) + \beta\, d(x,y)^k \right] \mathrm{d}y = \lambda \tag{21}$$

for all $x \in \mathrm{supp}(\rho^*)$.

Absorbing the factor of 2 into λ, we obtain the Euler–Lagrange condition:

$$\boxed{\int_\Omega \rho(y) \left[\alpha\, d(x,y) + \beta\, d(x,y)^k \right] \mathrm{d}y = \lambda \qquad \text{for } x \in \mathrm{supp}(\rho^*)} \tag{22}$$

Remark 2. *Equation* **??** *states that the* marginal communication cost *is constant over the support of the minimizing architecture.*

7 Regime Classification

Define the dimensionless architectural ratio

$$\chi := \frac{\alpha}{\beta}. \tag{23}$$

The value of χ determines which term dominates **??**.

7.1 Convergent Regime

If

$$\alpha \gg \beta, \tag{24}$$

then

$$\mathcal{T}[\rho] \approx \alpha \iint \rho(x)\rho(y) d(x,y)\ \mathrm{d}x\ \mathrm{d}y. \tag{25}$$

Minimization drives the support of ρ^* into a compact region:

$$\mathrm{diam}(\mathrm{supp}(\rho^*)) \to 0 \tag{26}$$

subject to **??** and thermal or fabrication bounds. This corresponds to a *convergent* architecture.

7.2 Diffuse Regime

If

$$\beta \gg \alpha, \tag{27}$$

then

$$\mathcal{T}[\rho] \approx \beta \iint \rho(x)\rho(y)d(x,y)^k \, \mathrm{d}x \, \mathrm{d}y. \tag{28}$$

In the idealized symmetric case, minimization favors a distributed profile:

$$\rho^*(x) \approx \frac{N}{|\Omega|}. \tag{29}$$

This corresponds to a *diffuse* architecture.

7.3 Hybrid Regime

If

$$\alpha \sim \beta, \tag{30}$$

neither term dominates, and the minimizer is generally nonuniform:

$$\rho^*(x) = \rho_{\text{core}}(x) + \rho_{\text{shell}}(x), \tag{31}$$

with higher density in the interior and distributed support outside. This yields a *hybrid* or core–shell architecture.

Theorem 2 (Computational Convergence Theorem)**.** *Under bounded clockspeed, distance-dependent bandwidth, and volumetric capacity scaling, the minimizer ρ^* of* **??** *converges to one of three qualitative architectural classes:*

(i) ***Convergent****, if $\alpha \gg \beta$,*

(ii) ***Diffuse****, if $\beta \gg \alpha$,*

(iii) ***Hybrid****, if $\alpha \sim \beta$.*

8 Geometric Interpretation

The theorem implies that architecture class is determined not by component labels (CPU, GPU, cache, interposer), but by the geometry induced by communication cost. In this view:

- **Convergent systems** prioritize low average distance,
- **Diffuse systems** prioritize high aggregate transport capacity,
- **Hybrid systems** trade off locality and distribution.

Thus the regime map is:

$$\boxed{\begin{cases} \chi \to \infty & \Rightarrow \text{Convergent} \\ \chi \approx 1 & \Rightarrow \text{Hybrid} \\ \chi \to 0 & \Rightarrow \text{Diffuse} \end{cases}} \tag{32}$$

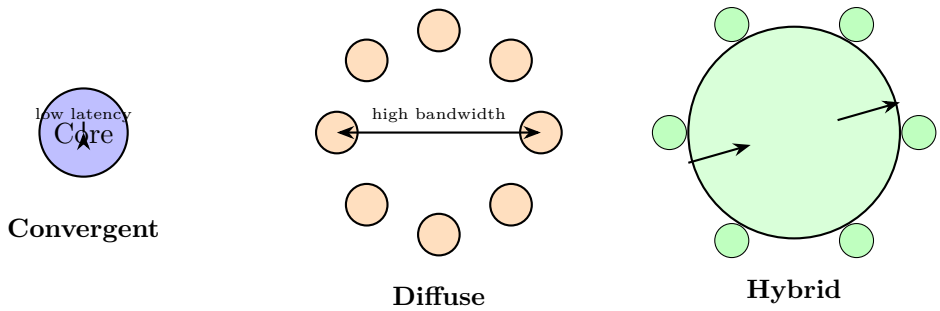

Figure 1: Qualitative architectural minimizers: convergent, diffuse, and hybrid.

9 Illustrative Geometry

10 Empirical Interpretation

The framework maps naturally onto observed system classes.

Table 1: Illustrative correspondence between predicted regimes and observed architectural forms.

System class	**Predicted regime**	**Observed spatial organization**
CPU cores and caches	Convergent	Compact compute clusters, deep cache hierarchy, strong locality optimization
GPU and TPU arrays	Diffuse	Wide spatial distribution of parallel units, throughput-first layout
Chiplet-based systems	Hybrid	Central logic plus distributed compute, memory, or I/O islands
HBM and stacked-memory systems	Hybrid	Memory placed in a shell-like arrangement around logic with short, wide links
3D logic/cache stacking	Convergent / Hybrid	Vertical compression of distance to preserve locality under area constraints

11 Falsifiability

The model is falsifiable because it predicts a correspondence between architectural objective and spatial organization:

- latency-dominated systems should converge spatially,
- throughput-dominated systems should diffuse spatially,
- mixed-objective systems should develop core–shell or modular hybrids.

A robust counterexample—for instance, a latency-first system consistently exhibiting diffuse optimal geometry, or a throughput-first system consistently exhibiting convergent geometry under comparable physical constraints—would count against the theory.

12 Design Implications

The framework predicts:

(1) increased 3D integration to reduce average communication distance,

(2) more near-memory and in-memory compute,

(3) growth of heterogeneous core–shell systems,

(4) software and runtimes that map workloads based on their effective α/β ratio.

As frequency saturates, architectural advantage shifts toward minimizing spatial communication cost:

$$\boxed{\min_{\rho} \iint_{\Omega\times\Omega} \rho(x)\rho(y)\left[\alpha d(x,y) + \beta d(x,y)^k\right] \mathrm{d}x\ \mathrm{d}y \quad \text{subject to} \quad f(t) \le \min\left(Ke^{-\gamma t}, \frac{v}{L}\right)} \tag{33}$$

13 Conclusion

We have presented a geometric framework for compute architecture grounded in three physical constraints: bounded clockspeed, distance-dependent bandwidth, and volumetric capacity scaling. Under these assumptions, the design of architecture becomes a constrained minimization problem over space. Its solutions fall into convergent, diffuse, and hybrid regimes, providing a unified interpretation of modern CPUs, GPUs, chiplets, and stacked-memory systems.

Final insight. When frequency scaling stops, geometry becomes the architecture.